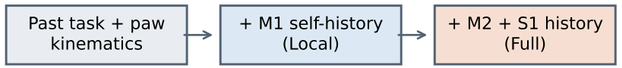
Recorded M2/S1 history predicts M1 at both events but is not selectively greater during grasp

a A direct test of an author-stated grasp hypothesis



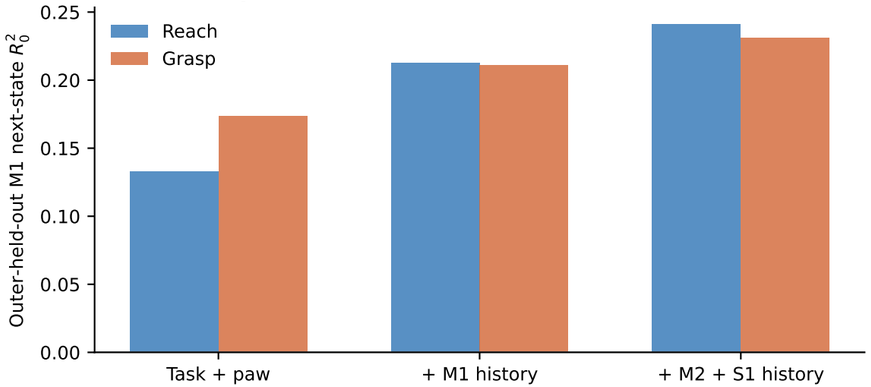
Target M1 population state in the next 50 ms

Primary contrast

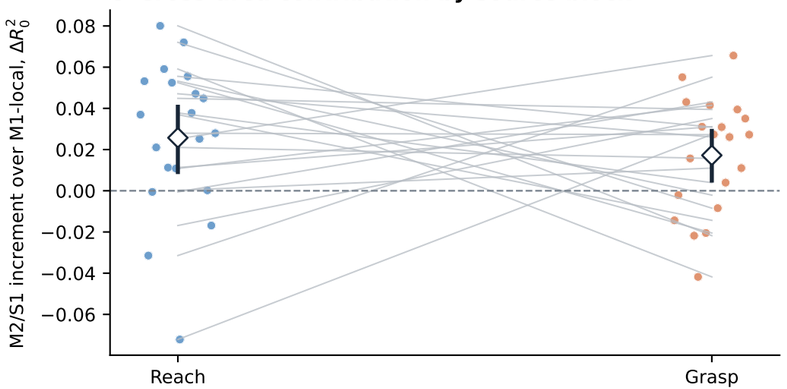
(Full − Local)grasp − (Full − Local)reach

Positive supports greater recorded cross-area predictive contribution around grasp—not causal or anatomical input.

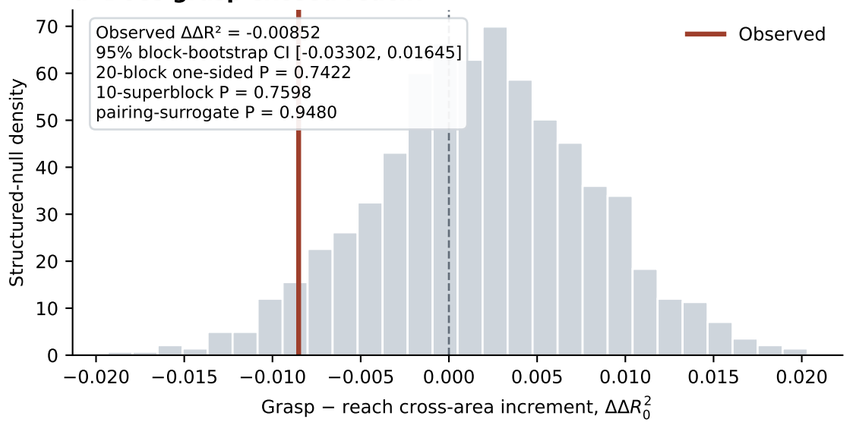
b Held-out prediction from nested models



c Cross-area contribution by source block



d Does grasp exceed reach?



One mouse-day; strict-QC M1/M2/S1 spikes; 75 matched trials × 5 cutoffs/event. Complete-trial chronological folds; every neural transform is outer-training-only. Blocks quantify repeated time within one recording, not animal-level replication. Predictive contribution is not anatomical input, information flow or causality.